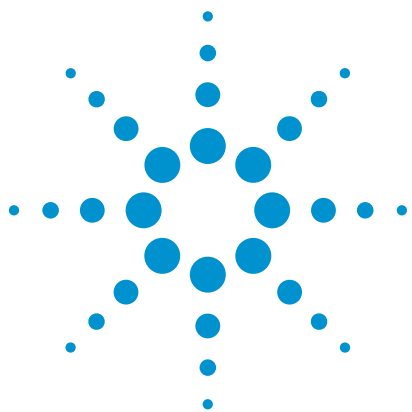
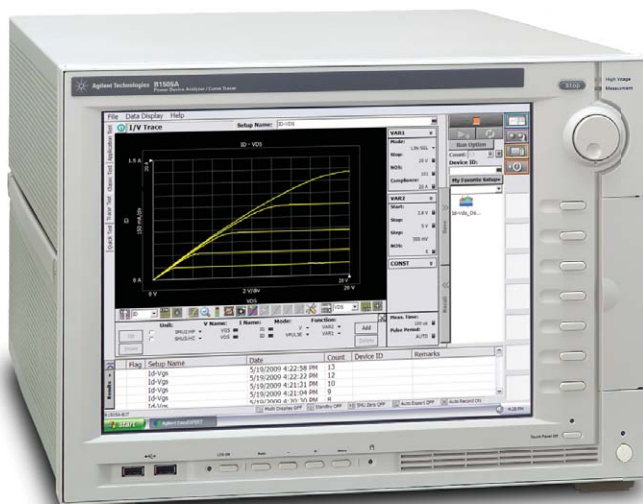




Agilent B1505A

Power Device Analyzer/Curve Tracer

Data Sheet



*Agilent B1505A
Power Device
Analyzer/Curve
Tracer*

Introduction

The B1505A Power Device Analyzer/ Curve Tracer is a single-box solution with next-generation curve tracer functionality that can accurately evaluate and characterize power devices at up to 3000 volts and 20 amps. It can also perform capacitance measurements at high voltage biases (up to 3000 V). The B1505A with EasyEXPERT software includes a curve tracer mode that combines

familiar curve tracer functionality with the convenience of a PC-based instrument; this makes it easy for traditional curve-tracer users to become productive quickly. The net result is improved ease of use, better data analysis and simplified data management for the measurement of power devices and power circuitry.



Agilent Technologies

Basic features

- Performs IV measurements (3000 V, 20 A)
- Performs high bias voltage CV measurements
- Easy to use EasyEXPERT test environment
- Curve tracer test mode
- Modular configuration with ten module slots for supported modules
- Multiple SMU types available: HPSMU, HCSMU, and HVSMU
- Multi-frequency capacitance measurement unit (MFCMU) (1 kHz to 5 MHz) available
- Standard accessories for package test and wafer test: test fixture, module selector, and high voltage bias-tee
- 4.2-amp ground unit included standard with the mainframe
- GPIB port for instrument control
- Self-test, self-calibration, diagnostics

Specification conditions

The measurement and output accuracy are specified under the conditions listed below. Note: The SMU measurement and output accuracies are specified at the SMU connector terminals, using the Zero Check terminal as a reference.

1. Temperature: $23\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$
2. Humidity: 20% to 70%
3. Self-calibration after a 40 minute warm-up is required.
4. Ambient temperature change less than $\pm 1\text{ }^{\circ}\text{C}$ after self-calibration execution. (Note: This does not apply to the MFCMU).
5. Measurement made within one hour after self-calibration execution. (Note: This does not apply to the MFCMU).
6. Calibration period: 1 year
7. SMU integration time setting:
1 PLC (1 nA to 1 A range, voltage range), 200 μs (20 A range)
Averaging of high-speed ADC:
128 samples per 1 PLC
8. SMU filter: ON (for HPSMU)
9. SMU measurement terminal connection: Kelvin connection (for HPSMU, HCSMU), non-Kelvin (for HVSMU)

Note: This document lists specifications and supplemental characteristics for the B1505A and its associated modules. The specifications are the standards against which the B1505A and its associated modules are tested. When the B1505A or any of its associated modules are shipped from the factory, they meet the specifications. The “supplemental” characteristics described in the following specifications are not guaranteed, but provide useful information about the functions and performance of the instrument.

Note: When you install or exchange modules in the B1505A mainframe, Agilent can guarantee that the modules will function and that the module design will perform to its specifications. However, if you require a guarantee that the modules meet their specifications then you must perform a calibration on the instrument.

B1505A Specifications

Supported plug-In modules

The B1505A supports ten slots for plug-in modules.

Part number	Description	Slots occupied	Range of operation	Measure resolution
B1510A	High Power Source Monitor Unit (HPSMU)	2	-200 V to 200 V, -1 A to 1 A	2 μ V, 10 fA
B1512A	High Current Source Monitor Unit (HCSMU)	2	-40 V to 40 V, -1 A to 1 A -20 V to 20 V, -20 A to 20 A (Pulse only)	200 nV, 10 pA
B1513A	High Voltage Source Monitor Unit (HVSMU)	2	-3000 V to 3000 V, -4 mA to 4 mA -1500 V to 1500 V, -8 mA to 8 mA	200 μ V, 10 fA
B1520A	Multi Frequency Capacitance Measurement Unit (MFCMU)	1	1 kHz to 5 MHz	0.035 fFrms ¹

1. Valid when connecting a 10 pF capacitor to the measurement terminals under the following measurement conditions: a frequency of 1 MHz, a signal level of 250 mV AC, and a measurement time of 1 PLC. The display resolution is 0.000001 fF at 1 fF order by 6 digits display.

Maximum module configuration

The total power consumption of all modules cannot exceed 84 W. Under this rule, the B1505A can contain up to 2 dual-slot HPSMUs, up to 2 dual-slot HCSMUs, and up to 1 dual-slot HVSMU; it can contain any combination of these SMUs. Up to 1 single-slot MFCMU can be installed per B1505A mainframe.

The installation order of the modules is: HPSMU, MFCMU, HCSMU, and HVSMU starting from the bottom of B1505A mainframe.

Maximum voltage between Common and Ground

$\leq \pm 42$ V

Ground unit (GNDU) specifications

The GNDU is furnished with the B1505A mainframe.

Output voltage: 0 V $\pm$ 100 μ V

Maximum sink current: $\pm$ 4.2 A

Output terminal/connection:

Triaxial connector, Kelvin
(remote sensing)

GNDU supplemental characteristics

Load capacitance: 1 μ F

Cable resistance:

For $I_s \leq 1.6$ A: Force line R < 1 Ω

For 1.6 A < $I_s \leq 2.0$ A: Force line R < 0.7 Ω

For 2.0 A < $I_s \leq 4.2$ A: Force line R < 0.35 Ω

For all cases: Sense line R $\leq$ 10 Ω

Where I_s is the current being sunk by the GNDU.

HPSMU Module Specifications

Voltage range, resolution, and accuracy (high resolution ADC)

Voltage range	Force resolution	Measure resolution	Force accuracy ¹ ±(% + mV)	Measure accuracy ¹ ±(% + mV)	Maximum current
±2 V	100 µV	2 µV	±(0.018 + 0.4)	±(0.01 + 0.14)	1 A
±20 V	1 mV	20 µV	±(0.018 + 3)	±(0.01 + 0.14)	1 A
±40 V	2 mV	40 µV	±(0.018 + 6)	±(0.01 + 1)	500 mA
±100 V	5 mV	100 µV	±(0.018 + 15)	±(0.012 + 2.5)	125 mA
±200 V	10 mV	200 µV	±(0.018 + 30)	±(0.014 + 2.8)	50 mA

1. ± (% of reading value + offset value in mV)

Current range, resolution, and accuracy (high resolution ADC)

Current range	Force resolution	Measure resolution	Force accuracy ¹ ±(% + A + A)	Measure accuracy ¹ ±(% + A + A)	Maximum voltage
±1 nA	50 fA	10 fA	±(0.1 + 3E-13 + Vo x 1E-15)	±(0.1 + 3E-13 + Vo x 1E-15)	200 V
±10 nA	500 fA	10 fA	±(0.1 + 3E-12 + Vo x 1E-14)	±(0.1 + 25E-13 + Vo x 1E-14)	200 V
±100 nA	5 pA	100 fA	±(0.05 + 3E-11 + Vo x 1E-13)	±(0.05 + 25E-12 + Vo x 1E-13)	200 V
±1 µA	50 pA	1 pA	±(0.05 + 3E-10 + Vo x 1E-12)	±(0.05 + 1E-10 + Vo x 1E-12)	200 V
±10 µA	500 pA	10 pA	±(0.05 + 3E-9 + Vo x 1E-11)	±(0.04 + 2E-9 + Vo x 1E-11)	200 V
±100 µA	5 nA	100 pA	±(0.035 + 15E-9 + Vo x 1E-10)	±(0.03 + 3E-9 + Vo x 1E-10)	200 V
±1 mA	50 nA	1 nA	±(0.04 + 15E-8 + Vo x 1E-9)	±(0.03 + 6E-8 + Vo x 1E-9)	200 V
±10 mA	500 nA	10 nA	±(0.04 + 15E-7 + Vo x 1E-8)	±(0.03 + 2E-7 + Vo x 1E-8)	200 V
±100 mA	5 µA	100 nA	±(0.045 + 15E-6 + Vo x 1E-7)	±(0.04 + 6E-6 + Vo x 1E-7)	200 V ²
±1 A	50 µA	1 µA	±(0.4 + 3E-4 + Vo x 1E-6)	±(0.4 + 15E-5 + Vo x 1E-6)	200 V ²

1. ± (% of reading value + fixed offset in A + proportional offset in A), Vo is the output voltage in V.)

2. 200 V (Io ≤ 50 mA), 100 V (50 mA < Io ≤ 125 mA), 40 V (125 mA < Io ≤ 500 mA), 20 V (500 mA < Io ≤ 1 A), Io is the output current in Amps.

Voltage range, resolution, and accuracy (high speed ADC)

Voltage range	Force resolution	Measure resolution	Force accuracy ¹ ±(% + mV)	Measure accuracy ¹ ±(% + mV)	Maximum current
±2 V	100 µV	100 µV	±(0.018 + 0.4)	±(0.01 + 0.7)	1 A
±20 V	1 mV	1 mV	±(0.018 + 3)	±(0.01 + 4)	1 A
±40 V	2 mV	2 mV	±(0.018 + 6)	±(0.015 + 8)	500 mA
±100 V	5 mV	5 mV	±(0.018 + 15)	±(0.02 + 20)	125 mA
±200 V	10 mV	10 mV	±(0.018 + 30)	±(0.035 + 40)	50 mA

1. ±(% of reading value + offset value in mV). Averaging is 128 samples in 1 PLC.

Current range, resolution, and accuracy (high speed ADC)

Current range	Force resolution	Measure resolution	Force accuracy ¹ ±(% + A + A)	Measure accuracy ¹ ±(% + A + A)	Maximum voltage
±1 nA	50 fA	50 fA	±(0.1 + 3E-13 + Vo x 1E-15)	±(0.25 + 3E-13 + Vo x 1E-15)	200 V
±10 nA	500 fA	500 fA	±(0.1 + 3E-12 + Vo x 1E-14)	±(0.25 + 2E-12 + Vo x 1E-14)	200 V
±100 nA	5 pA	5 pA	±(0.05 + 3E-11 + Vo x 1E-13)	±(0.1 + 2E-11 + Vo x 1E-13)	200 V
±1 µA	50 pA	50 pA	±(0.05 + 3E-10 + Vo x 1E-12)	±(0.1 + 2E-10 + Vo x 1E-12)	200 V
±10 µA	500 pA	500 pA	±(0.05 + 3E-9 + Vo x 1E-11)	±(0.05 + 2E-9 + Vo x 1E-11)	200 V
±100 µA	5 nA	5 nA	±(0.035 + 15E-9 + Vo x 1E-10)	±(0.05 + 2E-8 + Vo x 1E-10)	200 V
±1 mA	50 nA	50 nA	±(0.04 + 15E-8 + Vo x 1E-9)	±(0.04 + 2E-7 + Vo x 1E-9)	200 V
±10 mA	500 nA	500 nA	±(0.04 + 15E-7 + Vo x 1E-8)	±(0.04 + 2E-6 + Vo x 1E-8)	200 V
±100 mA	5 µA	5 µA	±(0.045 + 15E-6 + Vo x 1E-7)	±(0.1 + 2E-5 + Vo x 1E-7)	200 V ²
±1 A	50 µA	50 µA	±(0.4 + 3E-4 + Vo x 1E-6)	±(0.5 + 3E-4 + Vo x 1E-6)	200 V ²

1. ±(% of reading value + fixed offset in A + proportional offset in A), Vo is the output voltage in V.)

2. 200 V (Io ≤ 50 mA), 100 V (50 mA < Io ≤ 125 mA), 40 V (125 mA < Io ≤ 500 mA), 20 V (500 mA < Io ≤ 1 A), Io is the output current in Amps.

Power consumption

Voltage source mode:

Voltage range	Power
2 V	20 x Ic (W)
20 V	20 x Ic (W)
40 V	40 x Ic (W)
100 V	100 x Ic (W)
200 V	200 x Ic (W)

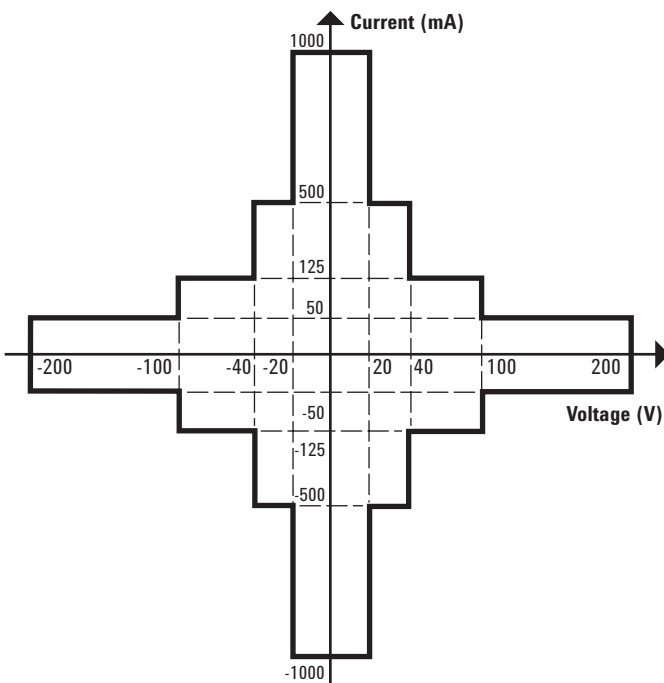
Where I_c is the current compliance setting.

Current source mode:

Voltage compliance	Power
$V_C \leq 20$	$20 \times I_o$ (W)
$20 < V_C \leq 40$	$40 \times I_o$ (W)
$40 < V_C \leq 100$	$100 \times I_o$ (W)
$100 < V_C \leq 200$	$200 \times I_o$ (W)

Where V_c is the voltage compliance setting and I_o is output current.

HPSMU measurement and output range



HCSMU Module Specifications

Voltage range, resolution, and accuracy

Voltage range	Force resolution	Measure resolution	Force accuracy ¹ $\pm(\% + \text{mV} + \text{mV})$	Measure accuracy ¹ $(\% + \text{mV} + \text{mV})$	Maximum current
$\pm 0.2 \text{ V}$	200 nV	200 nV	$\pm(0.06 + 0.14 + I_o \times 0.05)$	$\pm(0.06 + 0.14 + I_o \times 0.05)$	20 A
$\pm 2 \text{ V}$	2 μV	2 μV	$\pm(0.06 + 0.6 + I_o \times 0.5)$	$\pm(0.06 + 0.6 + I_o \times 0.5)$	20 A
$\pm 20 \text{ V}$	20 μV	20 μV	$\pm(0.06 + 3 + I_o \times 5)$	$\pm(0.06 + 3 + I_o \times 5)$	20 A
$\pm 40 \text{ V}$	40 μV	40 μV	$\pm(0.06 + 3 + I_o \times 10)$	$\pm(0.06 + 3 + I_o \times 10)$	1 A

1. $\pm(\% \text{ of reading value} + \text{offset voltage } V)$

Current range, resolution, and accuracy

Current range	Force resolution	Measure resolution	Force accuracy ¹ $(\% + A + A)$	Measure accuracy ¹ $(\% + A + A)$	Maximum voltage
$\pm 10 \mu\text{A}$	10 pA	10 pA	$\pm(0.06 + 2\text{E-}9 + V_o \times 1\text{E-}10)$	$\pm(0.06 + 2\text{E-}9 + V_o \times 1\text{E-}10)$	40 V
$\pm 100 \mu\text{A}$	100 pA	100 pA	$\pm(0.06 + 2\text{E-}8 + V_o \times 1\text{E-}9)$	$\pm(0.06 + 2\text{E-}8 + V_o \times 1\text{E-}9)$	40 V
$\pm 1 \text{ mA}$	1 nA	1 nA	$\pm(0.06 + 2\text{E-}7 + V_o \times 1\text{E-}8)$	$\pm(0.06 + 2\text{E-}7 + V_o \times 1\text{E-}8)$	40 V
$\pm 10 \text{ mA}$	10 nA	10 nA	$\pm(0.06 + 2\text{E-}6 + V_o \times 1\text{E-}7)$	$\pm(0.06 + 2\text{E-}6 + V_o \times 1\text{E-}7)$	40 V
$\pm 100 \text{ mA}$	100 nA	100 nA	$\pm(0.06 + 2\text{E-}5 + V_o \times 1\text{E-}6)$	$\pm(0.06 + 2\text{E-}5 + V_o \times 1\text{E-}6)$	40 V
$\pm 1 \text{ A}$	1 μA	1 μA	$\pm(0.4 + 2\text{E-}4 + V_o \times 1\text{E-}5)$	$\pm(0.4 + 2\text{E-}4 + V_o \times 1\text{E-}5)$	40 V
$\pm 20 \text{ A}^2$	20 μA	20 μA	$\pm(0.4 + 2\text{E-}3 + V_o \times 1\text{E-}4)$	$\pm(0.4 + 2\text{E-}3 + V_o \times 1\text{E-}4)$	20 V

1. $\pm(\% \text{ of reading value} + \text{fixed offset in } A + \text{proportional offset in } A)$, V_o is the output voltage in V.

2. Pulse mode only.

Power consumption

Voltage source mode:

Voltage range	Power
0.2 V	$40 \times I_c \text{ (W)}$
2 V	$40 \times I_c \text{ (W)}$
40 V	$40 \times I_c \text{ (W)}$

Where I_c is the current compliance setting.

For pulse current, $I_c = (\text{duty}) \times I_{\text{pulse}}$

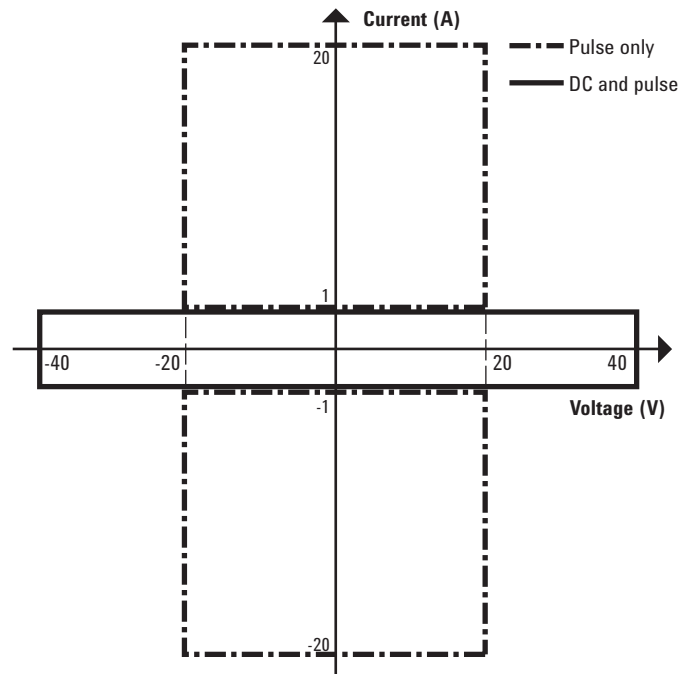
Current source mode:

Voltage compliance	Power
$V_c \leq 0.2$	$40 \times I_o \text{ (W)}$
$0.2 < V_c \leq 2$	$40 \times I_o \text{ (W)}$
$2 < V_c \leq 40$	$40 \times I_o \text{ (W)}$

Where V_c is the voltage compliance setting and I_o is output current.

For pulse current, $I_o = (\text{duty}) \times I_{\text{pulse}}$

HCSMU measurement and output range



HVSMU Module Specifications

Voltage range, resolution, and accuracy

Voltage range	Force resolution	Measure resolution	Force accuracy ¹ $\pm(\% + \text{mV})$	Measure accuracy ¹ $\pm(\% + \text{mV})$	Maximum current
$\pm 200 \text{ V}$	$200 \mu\text{V}$	$200 \mu\text{V}$	$\pm(0.03 + 40)$	$\pm(0.03 + 40)$	8 mA
$\pm 500 \text{ V}$	$500 \mu\text{V}$	$500 \mu\text{V}$	$\pm(0.03 + 100)$	$\pm(0.03 + 100)$	8 mA
$\pm 1500 \text{ V}$	1.5 mV	1.5 mV	$\pm(0.03 + 300)$	$\pm(0.03 + 300)$	8 mA
$\pm 3000 \text{ V}$	3 mV	3 mV	$\pm(0.03 + 600)$	$\pm(0.03 + 600)$	4 mA

1. $\pm(\% \text{ of reading value} + \text{offset voltage } V)$

Current range, resolution, and accuracy

Current range	Force resolution	Measure resolution	Force accuracy ¹ $\pm(\% + A + A)$	Measure accuracy ¹ $\pm(\% + A + A)$	Maximum voltage	Minimum set current ²
$\pm 1 \text{ nA}$	10 fA	10 fA	$\pm(0.1 + 6\text{E-}13 + V_o \times 1\text{E-}15)$	$\pm(0.1 + 6\text{E-}13 + V_o \times 1\text{E-}15)$	3000 V	1 pA
$\pm 10 \text{ nA}$	100 fA	100 fA	$\pm(0.1 + 25\text{E-}13 + V_o \times 1\text{E-}15)$	$\pm(0.1 + 25\text{E-}13 + V_o \times 1\text{E-}15)$	3000 V	1 pA
$\pm 100 \text{ nA}$	100 fA	100 fA	$\pm(0.05 + 25\text{E-}12 + V_o \times 1\text{E-}13)$	$\pm(0.05 + 25\text{E-}12 + V_o \times 1\text{E-}13)$	3000 V	100 pA
$\pm 1 \mu\text{A}$	1 pA	1 pA	$\pm(0.05 + 1\text{E-}10 + V_o \times 1\text{E-}13)$	$\pm(0.05 + 1\text{E-}10 + V_o \times 1\text{E-}13)$	3000 V	100 pA
$\pm 10 \mu\text{A}$	10 pA	10 pA	$\pm(0.04 + 2\text{E-}9 + V_o \times 1\text{E-}11)$	$\pm(0.04 + 2\text{E-}9 + V_o \times 1\text{E-}11)$	3000 V	10 nA
$\pm 100 \mu\text{A}$	100 pA	100 pA	$\pm(0.03 + 3\text{E-}9 + V_o \times 1\text{E-}11)$	$\pm(0.03 + 3\text{E-}9 + V_o \times 1\text{E-}11)$	3000 V	10 nA
$\pm 1 \text{ mA}$	1 nA	1 nA	$\pm(0.03 + 6\text{E-}8 + V_o \times 1\text{E-}10)$	$\pm(0.03 + 6\text{E-}8 + V_o \times 1\text{E-}10)$	3000 V	100 nA
$\pm 10 \text{ mA}$	10 nA	10 nA	$\pm(0.03 + 2\text{E-}7 + V_o \times 1\text{E-}9)$	$\pm(0.03 + 2\text{E-}7 + V_o \times 1\text{E-}9)$	1500 V	1 μA

1. $\pm(\% \text{ of reading value} + \text{fixed offset in } A + \text{proportional offset in } A)$, V_o is the output voltage in V.

2. Output current needs to be set more than current shown in the table.

HVSMU measurement and output range

Power consumption

Voltage source mode:

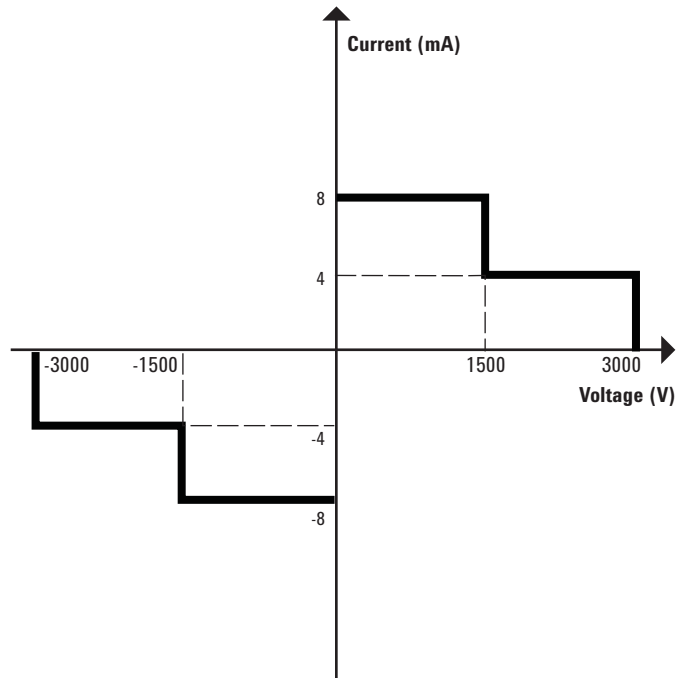
Current compliance	Power
$I_c \leq 4\text{m}$	$3000 \times I_c \text{ (W)}$
$4\text{m} < I_c \leq 8\text{m}$	$1500 \times I_c \text{ (W)}$

Where I_c is the current compliance setting.

Current source mode:

Voltage compliance	Power
$V_c \leq 1500$	$1500 \times I_o \text{ (W)}$
$1500 < V_c \leq 3000$	$3000 \times I_o \text{ (W)}$

Where V_c is the voltage compliance setting and I_o is output current.



Supplemental Characteristics

SMU source measurement mode

For HPSMU:

VFIM, IFVM

For HCSMU, HVSMU:

VFIM, VFVM, IFVM, IFIM

Output terminal/connection:

For HPSMU:

Dual triaxial connector,

Kelvin (remote sensing)

For HCSMU:

Triaxial connector (for sense) and
coaxial connector (for force)

Kelvin (remote sensing) by using
16493S-010 HCSMU Kelvin adapter

For HVSMU:

High voltage triaxial connector,
non-Kelvin

Voltage/current compliance (limiting)

The SMU can limit output voltage
or current to prevent damaging the
device under test.

Voltage:

0 V to ± 200 V (HPSMU)

0 V to ± 40 V (HCSMU)

0 V to ± 3000 V (HVSMU)

Current:

± 1 pA to ± 1 A (HPSMU)

± 10 nA to ± 20 A (HCSMU)

± 1 pA to ± 8 mA (HVSMU)

Compliance accuracy:

Same as the current or voltage set
accuracy.

Power compliance

For HPSMU:

Power: 0.001 to 20 W

Resolution: 0.001 W

For HCSMU:

Power: 0.001 to 40 W (DC)

0.001 to 400 W (Pulse)

Resolution: 0.001 W

For HVSMU:

No power compliance

SMU pulse measurement

Pulse width, period, and delay:

For HPSMU:

Pulse width: 500 μ s to 2 s

Pulse width resolution: 2 μ s

Pulse period: 5 ms to 5 s

Period $\geq$ delay + width + 2 ms

(when delay + width $\leq$ 100 ms)

Period $\geq$ delay + width + 10 ms

(when delay + width > 100 ms)

Pulse period resolution: 100 μ s

Pulse delay: 0 s

For HCSMU:

Pulse width:

50 μ s to 1 ms (20 A range)

50 μ s to 2 s (10 μ A to 1 A range)

Pulse width resolution: 2 μ s

Pulse period: 5 ms to 5 s

Pulse period resolution: 100 μ s

Pulse duty:

For 20 A range: $\leq 1\%$

For 10 μ A to 1 A range

Period $\geq$ delay + width + 2 ms

(when delay + width $\leq$ 100 ms)

Period $\geq$ delay + width + 10 ms

(when delay + width > 100 ms)

Pulse delay: 0 to (Period–width)

For HVSMU:

Pulse width: 500 μ s to 2 s

Pulse width resolution: 2 μ s

Pulse period: 5 ms to 5 s

Period $\geq$ delay + width + 2 ms

(when delay + width $\leq$ 100 ms)

Period $\geq$ delay + width + 10 ms

(when delay + width > 100 ms)

Pulse period resolution: 100 μ s

Pulse delay: 0 to (Period – width)

Pulse output limitation:

When the pulse voltage is more
than 1500 volts, the peak and
base of pulse should be same
polarities.

Pulse measurement delay:

2 μ s to (Period – pulse
measurement time – 2 m) s,
2 μ s resolution

Current compliance setting accuracy (for opposite polarity):

For HPSMU:

For 1 pA to 10 nA ranges:

V/I setting accuracy $\pm 12\%$ of
range

For 100 nA to 1 A ranges:

V/I setting accuracy $\pm 2.5\%$ of
range

For HCSMU:

For 10 μ A to 1 A ranges:

V/I setting accuracy $\pm 2.5\%$ of
range

For 20 A ranges:

V/I setting accuracy $\pm 0.6\%$ of
range

For HVSMU:

For 1 nA to 10 nA ranges:

V/I setting accuracy $\pm 12\%$ of
range

For 100 nA to 10 mA ranges:

V/I setting accuracy $\pm 2.5\%$ of
range

SMU pulse setting accuracy (fixed measurement range):

For HPSMU:

Width: $\pm 0.5\% \pm 50$ μ s

Period: $\pm 0.5\% \pm 100$ μ s

For HCSMU:

Width: $\pm 0.1\% \pm 2$ μ s

Period: $\pm 0.1\% \pm 100$ μ s

For HVSMU:

Width: $\pm 0.1\% \pm 2$ μ s

Period: $\pm 0.5\% \pm 100$ μ s

Minimum pulse measurement time:

16 μ s (HPSMU)

2 μ s (HCSMU)

6 μ s (HVSMU)

Voltage source output resistance:

(Force line, non-Kelvin connection)

0.2 Ω (HPSMU)

3 Ω (HVSMU, at 10 mA range)

Voltage measurement input resistance:

$\geq 10^{13}$ Ω (HPSMU)

$\geq 10^9$ Ω (HCSMU, ≤ 1 A),

80 k Ω (HCSMU, 20 A)

$\geq 10^{12}$ Ω (HVSMU)

Current source output resistance:

- $\geq 10^{13} \Omega$ (HPSMU)
- $\geq 10^9 \Omega$ (HCSMU, ≤ 1 A),
80 k Ω (HCSMU, 20 A)
- $\geq 10^{12} \Omega$ (HVSMU, at 10 nA range)

Maximum allowable cable resistance:

(Kelvin connection)

For HPSMU:

- Sense: 10 Ω
- Force: 10 Ω (≤ 100 mA),
1.5 Ω (>100 mA)

For HCSMU:

- Sense: 10 Ω
- Force: 0.6 Ω
(with Low Force (shield))

Maximum allowable inductance:

For HCSMU:

- Force 3 μ H
(with Low Force (shield))

Maximum load capacitance:

For HPSMU:

- 1 pA to 10 nA ranges: 1000 pF
- 100 nA to 10 mA ranges: 10 nF
- 100 mA and 1 A ranges: 100 μ F

For HCSMU: 12 nF

For HVSMU:

- 1 nA to 1 μ A ranges: 1000 pF
- 10 μ A to 10 mA ranges: 10 nF

Maximum guard capacitance:

- 900 pF (HPSMU)
- 1500 pF (HVSMU)

Maximum shield capacitance:

- 5000 pF (HPSMU, HVSMU)

Noise characteristics:

For HPSMU, HVSMU

- Voltage source:
0.01% of V range (rms.)
- Current source:
0.1% of I range (rms.)

For HCSMU

- Voltage/Current source:
100 mV (0 to peak) max

Overshoot:

For HPSMU

- Voltage source: 0.03% of V range
- Current source: 1% of I range

For HCSMU (filter ON)

- Voltage/Current source:
10% of range

For HVSMU

- Voltage source: 1V (resistive load)
- Current source: 1% of I range

Range switching transient noise:

For HPSMU (filter ON):

- Voltage ranging: 250 mV
- Current ranging: 70 mV

For HCSMU:

- 10 μ A to 1 A ranges:
Voltage ranging: 250 mV
Current ranging: 70 mV
- 20 A ranges:
Voltage ranging: 5 V max

For HVSMU:

- Voltage ranging: 300 mV
- Current ranging: 300 mV

Maximum guard offset voltage:

- ± 1 mV (HPSMU)
- ± 5 mV (HVSMU)

Maximum slew rate:

- 0.2 V/ μ s (HPSMU)
- 1 V/ μ s (HCSMU)
- 0.4 V/ μ s (HVSMU)

Output settling time

For HVSMU:

- Output settling time: 500 μ s
- To reach 0.01% of settling value.
- Conditions:
100 V step, 8 mA compliance,
1000 pF load capacitance

MFCMU (multi frequency capacitance measurement unit) module specifications

Measurement functions

Measurement parameters:

Cp-G, Cp-D, Cp-Q, Cp-Rp, Cs-Rs,
Cs-D, Cs-Q, Lp-G, Lp-D, Lp-Q, Lp-Rp,
Ls-Rs, Ls-D, Ls-Q, R-X, G-B, Z-θ, Y-θ

Ranging:

Auto and fixed

Measurement terminal:

Four-terminal pair configuration,
four BNC (female) connectors

Cable length:

1.5 m or 3 m, automatic
identification of accessories

Test signal

Frequency:

Range: 1 kHz to 5 MHz
Resolution: 1 mHz (minimum)
Accuracy: $\pm 0.008\%$

Output signal level:

Range: 10 mV_{rms} to 250 mV_{rms}
Resolution: 1 mV_{rms}
Accuracy:

$\pm(10.0\% + 1 \text{ mV}_{\text{rms}})$ at the
measurement port of the MFCMU
 $\pm(15.0\% + 1 \text{ mV}_{\text{rms}})$ at the
measurement port of MFCMU
cable (1.5 m or 3.0 m)

Output impedance: 50 Ω, typical

Signal level monitor:

Range: 10 mVrms to 250 mV_{rms}

Accuracy:

$\pm(10.0\% \text{ of reading} + 1 \text{ mV}_{\text{rms}})$
at the measurement port of the
MFCMU
 $\pm(15.0\% + 1 \text{ mV}_{\text{rms}})$
at the measurement port of
MFCMU cable (1.5 m or 3.0 m)

DC bias function

DC bias:

Range: 0 to $\pm 25 \text{ V}$
Resolution: 1 mV
Accuracy: $\pm(0.5\% + 5.0 \text{ mV})$
at the measurement port or the
MFCMU or the MFCMU cable
(1.5 m/3 m)

Maximum DC bias current

(Supplemental characteristics):

Impedance measurement range	Maximum DC bias current
50 Ω	10 mA
100 Ω	10 mA
300 Ω	10 mA
1 kΩ	1 mA
3 kΩ	1 mA
10 kΩ	100 μA
30 kΩ	100 μA
100 kΩ	10 μA
300 kΩ	10 μA

Output impedance: 50 Ω, typical

DC bias monitor:

Range: 0 to $\pm 25 \text{ V}$
Accuracy (open load):
 $\pm(0.2\% \text{ of reading} + 10.0 \text{ mV})$
at the measurement port or the
MFCMU cable (1.5 m/3 m)

Sweep characteristics

Available sweep parameters:

Oscillator level, DC bias voltage,
frequency

Sweep type: linear, log

Sweep mode: single, double

Sweep direction: up, down

Number of measurement points:

Maximum 1001 points

Measurement accuracy

The following parameters are used to
express the impedance measurement
accuracy at the measurement port of
the MFCMU or the MFCMU cable
(1.5 m or 3 m).

Z_x : Impedance measurement value (Ω)

D_x : Measurement value of D

$$E = E_p' + (Z_s' / |Z_x| + Y_0' |Z_x|) \times 100 (\%)$$

$$E_p' = E_{PL} + E_{POSC} + E_p (\%)$$

$$Y_0' = Y_{OL} + Y_{OSC} + Y_0 (S)$$

$$Z_s' = Z_{SL} + Z_{OSC} + Z_s (\Omega)$$

|Z| accuracy
 $\pm E (\%)$

θ accuracy
 $\pm E / 100 (\text{rad})$

C accuracy
at $D_x \leq 0.1$
 $\pm E (\%)$

$$\text{at } D_x > 0.1 \\ \pm E \times \sqrt{(1 + D_x^2)} (\%)$$

D accuracy
at $D_x \leq 0.1$
 $\pm E / 100$

$$\text{at } D_x > 0.1 \\ \pm E \times (1 + D_x) / 100$$

G accuracy
at $D_x \leq 0.1$
 $\pm E / D_x (\%)$

$$\text{at } D_x > 0.1 \\ \pm E \times \sqrt{(1 + D_x^2)} / D_x (\%)$$

Note: measurement accuracy is speci-
fied under the following conditions:

Temperature: 23 °C ± 5 °C

Integration time: 1 PLC

Parameters E_{POSC} Z_{OSC}		
Oscillator level	E_{POSC} (%)	Z_{OSC} (m Ω)
$125 \text{ mV} < V_{\text{OSC}} \leq 250 \text{ mV}$	$0.03 \times (250 / V_{\text{OSC}} - 1)$	$5 \times (250 / V_{\text{OSC}} - 1)$
$64 \text{ mV} < V_{\text{OSC}} \leq 125 \text{ mV}$	$0.03 \times (125 / V_{\text{OSC}} - 1)$	$5 \times (125 / V_{\text{OSC}} - 1)$
$32 \text{ mV} < V_{\text{OSC}} \leq 64 \text{ mV}$	$0.03 \times (64 / V_{\text{OSC}} - 1)$	$5 \times (64 / V_{\text{OSC}} - 1)$
$V_{\text{OSC}} \leq 32 \text{ mV}$	$0.03 \times (32 / V_{\text{OSC}} - 1)$	$5 \times (64 / V_{\text{OSC}} - 1)$

V_{OSC} is oscillator level in mV.

Parameters E_{PL} Y_{OL} Z_{SL}			
Cable length	E_{PL} (%)	Y_{OL} (nS)	Z_{SL} (m Ω)
1.5 m	$0.02 + 3 \times f / 100$	$750 \times f / 100$	5.0
3 m	$0.02 + 5 \times f / 100$	$1500 \times f / 100$	5.0

f is frequency in MHz. If measurement cable is extended, open compensation, short compensation, and load compensation must be performed.

Parameters Y_{OSC} Y_0 E_p Z_s				
Frequency	Y_{OSC} (nS)	Y_0 (nS)	E_p (%)	Z_s (m Ω)
$1 \text{ kHz} \leq f \leq 200 \text{ kHz}$	$1 \times (125 / V_{\text{OSC}} - 0.5)$	1.5	0.095	5.0
$200 \text{ kHz} < f \leq 1 \text{ MHz}$	$2 \times (125 / V_{\text{OSC}} - 0.5)$	3.0	0.095	5.0
$1 \text{ MHz} < f \leq 2 \text{ MHz}$	$2 \times (125 / V_{\text{OSC}} - 0.5)$	3.0	0.28	5.0
$2 \text{ MHz} < f$	$20 \times (125 / V_{\text{OSC}} - 0.5)$	30.0	0.28	5.0

f is frequency in Hz.

V_{OSC} is oscillator level in mV.

Example of calculated C/G measurement accuracy				
Frequency	Measured capacitance	C accuracy ¹	Measured conductance	G accuracy ¹
5 MHz	1 pF	$\pm 0.61\%$	$\leq 3 \mu\text{S}$	$\pm 192 \text{ nS}$
	10 pF	$\pm 0.32\%$	$\leq 31 \mu\text{S}$	$\pm 990 \text{ nS}$
	100 pF	$\pm 0.29\%$	$\leq 314 \mu\text{S}$	$\pm 9 \mu\text{S}$
	1 nF	$\pm 0.32\%$	$\leq 3 \text{ mS}$	$\pm 99 \mu\text{S}$
1 MHz	1 pF	$\pm 0.26\%$	$\leq 628 \text{ nS}$	$\pm 16 \text{ nS}$
	10 pF	$\pm 0.11\%$	$\leq 6 \mu\text{S}$	$\pm 71 \text{ nS}$
	100 pF	$\pm 0.10\%$	$\leq 63 \mu\text{S}$	$\pm 624 \text{ nS}$
	1 nF	$\pm 0.10\%$	$\leq 628 \mu\text{S}$	$\pm 7 \mu\text{S}$
100 kHz	10 pF	$\pm 0.18\%$	$\leq 628 \text{ nS}$	$\pm 11 \text{ nS}$
	100 pF	$\pm 0.11\%$	$\leq 6 \mu\text{S}$	$\pm 66 \text{ nS}$
	1 nF	$\pm 0.10\%$	$\leq 63 \mu\text{S}$	$\pm 619 \text{ nS}$
	10 nF	$\pm 0.10\%$	$\leq 628 \mu\text{S}$	$\pm 7 \mu\text{S}$
10 kHz	100 pF	$\pm 0.18\%$	$\leq 628 \text{ nS}$	$\pm 11 \text{ nS}$
	1 nF	$\pm 0.11\%$	$\leq 6 \mu\text{S}$	$\pm 66 \text{ nS}$
	10 nF	$\pm 0.10\%$	$\leq 63 \mu\text{S}$	$\pm 619 \text{ nS}$
	100 nF	$\pm 0.10\%$	$\leq 628 \mu\text{S}$	$\pm 7 \mu\text{S}$
1 kHz	100 pF	$\pm 0.92\%$	$\leq 63 \text{ nS}$	$\pm 6 \text{ nS}$
	1 nF	$\pm 0.18\%$	$\leq 628 \text{ nS}$	$\pm 11 \text{ nS}$
	10 nF	$\pm 0.11\%$	$\leq 6 \mu\text{S}$	$\pm 66 \text{ nS}$
	100 nF	$\pm 0.10\%$	$\leq 63 \mu\text{S}$	$\pm 619 \text{ nS}$

1. The capacitance and conductance measurement accuracy is specified under the following conditions:

$D_x \leq 0.1$

Integration time: 1 PLC

Test signal level: 30 mV_{rms}

At four-terminal pair port of MFCMU

Accessories

N1258A module selector

Specifications

Input terminals:

HPSMU force port, 1 ea., (Triaxial)
HPSMU sense port, 1 ea., (Triaxial)
HCSMU force port, 1 ea. (BNC)
HCSMU sense port, 1 ea. (Triaxial)
HVSMU port, 1 ea. (HV triaxial)
GNDU port, 1 ea. (Triaxial)
Digital I/O port, 1 ea. (D-sub 25 pin)
AC power line connector, 1 ea.

Output terminal:

High force (HV triaxial)
High sense (HV triaxial)
Low force (BNC)
Low sense (BNC)
External relay control output
(D-sub 25 pin)

Protection:

HPSMU, GNDU, HCSMU Low Force

Power indicator:

LED turns yellow when AC power is applied and turns green the module selector is ready to use.

Status indicator:

Green LED lights to indicate the present connection path of module selector; Open, HCSMU, HPSMU, or HVSMU.

Maximum voltage/current:

For HPSMU:
 ± 200 V/1 A
For HCSMU:
 ± 40 V/1 A, ± 20 V/20 A
(Pulse width 1 ms, duty 1%)
For HVSMU:
 ± 3000 V/4 mA, ± 1500 V/8 mA

Supplemental characteristics

Leakage current:

For HPSMU:
10 pA at 200 V
For HCSMU:
100 pA at 10 V (High Force to Low Force, High Sense to Low Sense)
For HVSMU:
10 pA at 1500 V (humidity range: 20% to 70% RH)
20 pA at 3000 V (humidity range: 20% to 50% RH)

N1259A test fixture

Specifications

Input terminals:

HPSMU port, 2 ea.
Force, sense (Triaxial)
HCSMU port, 2 ea.
Force (BNC), sense (Triaxial)
HVSMU port, 1 ea. (HV triaxial)
GNDU port, 1 ea. (Triaxial)
AUX port, 2 ea. (BNC)
Interlock port, 1 ea.

Protection:

HPSMU, GNDU, HCSMU Low Force terminal

High voltage indicator:

LED turns red when a SMU output voltage is over 42 V.

Maximum voltage/current:

For HPSMU:
Force: ± 200 V/1 A
Sense: ± 200 V
For HCSMU:
High Force: ± 40 V/1 A, ± 20 V/20 A (Pulse width 1 ms, duty 1%)
Low Force: ± 40 V/1 A, ± 20 V/20 A (Pulse width 1 ms, duty 1%)
High Sense: ± 40 V
Low Sense: ± 40 V
For HVSMU:
Force: ± 3000 V/4 mA, ± 1500 V/8 mA

Note: The total power consumption of all modules cannot exceed 50 W when using test fixture under the condition that operating temperature is more than 35 °C.

Supplemental characteristics

Leakage current:

For HPSMU (Force, Sense):
10 pA at 200 V (Force, Sense)
For HCSMU (High Force, High sense):
100 pA at 10 V
For HVSMU (Force):
10 pA at 1500 V (humidity range: 20% to 70% RH)
20 pA at 3000 V (humidity range: 20% to 50% RH)

N1259A-010 inline package socket module (3 pin)

Specifications

Number of terminal:

Sockets, 6 ea. ($\varnothing 4$ mm jack (banana))

DUT interface:

Inline package socket (3-pin)

Maximum voltage for terminals:

3000 Vdc

N1259A-011 universal socket module

Specifications

Number of terminal:

Sockets, 8 ea. ($\varnothing 4$ mm jack (banana))

Maximum voltage for terminals:

3000 Vdc

N1259A-020 high voltage bias-tee

Specifications

Input terminals:

DC bias input, 1 ea.
($\varnothing 4$ mm jack (banana))
MFCMU port, 1 ea.
Hcur, Hpot, Lcur, Lpot, (BNC)

Output terminal:

MFCMU port
High (SHV)
Low (SHV)

External DC bias voltage: ± 3000 V

Frequency:

10 kHz to 1 MHz (150Ω at 10 kHz)

Series capacitance: 110 nF $\pm 5\%$

Input resistance: $100 \text{ k}\Omega \pm 1\%$

N1259A-021 1 M Ω resistor box

Specifications

Input/output terminals:
Ø4 mm jack (banana), 1 ea.
Resistance: 1 M Ω $\pm$ 5%
Maximum voltage: $\pm$ 3000 V
Power rating: 9 W

Supplemental characteristics

Leakage current: 10 pA at 100 V

N1259A-022 100 k Ω resistor box

Specifications

Input/output terminals:
Ø4 mm jack (banana), 1 ea.
Resistance: 100 k Ω $\pm$ 5%
Maximum voltage: $\pm$ 3000 V
Power rating: 6.4 W

Supplemental characteristics

Leakage current: 10 pA at 100 V

N1259A-030 1 k Ω resistor box for gate

Specifications

Input/output terminals:
Ø4 mm jack (banana), 1 ea.
Resistance: 1 k Ω $\pm$ 10%
Maximum voltage: $\pm$ 200 V
Maximum power: 1 W

Supplemental characteristics

Leakage current: 10 pA at 100 V

N1259A-300 module selector for test fixture

Specifications

Input terminals:
HPSMU port, 1 ea.
Force, sense (Triaxial)
HCSMU port, 1 ea.
Force (BNC), sense (Triaxial)
HVSMU port, 1 ea. (HV triaxial)
GNDU port, 1 ea. (Triaxial)
Digital I/O port, 1 ea. (D-sub 25 pin)
AC power line connector, 1 ea.

Output terminal:
High force and guard
High sense and guard
Low force
Low sense
(Ø4 mm jack (banana))

Protection:
HPSMU, GNDU, HCSMU Low Force

Power indicator:
LED turns yellow when AC power is applied and turns green the module selector is ready to use.

Status indicator:
Green LED lights to indicate the present connection path of module selector; Open, HCSMU, HPSMU, or HVSMU.

Maximum voltage/current:
For HPSMU:
 $\pm$ 200 V/1 A
For HCSMU:
 $\pm$ 40 V/1 A, $\pm$ 20 V/20 A
(Pulse width 1 ms, duty 1%)
For HVSMU:
 $\pm$ 3000 V/4 mA, $\pm$ 1500 V/8 mA

Supplemental characteristics

Leakage current:
For HPSMU:
10 pA at 200 V
For HCSMU:
100 pA at 10 V (High Force to Low Force, High Sense to Low Sense)
For HVSMU:
10 pA at 1500 V (humidity range: 20% to 70% RH)
30 pA at 3000 V (humidity range: 20% to 50% RH)

N1260A high voltage bias-tee

Specifications

Input terminals:
HVSMU port, 1 ea. (HV triaxial)
MFCMU port, 1 ea.
(4 BNC, Hp, Hc, Lp, Hc)
Output terminal:
H-AC Guard (SHV connector)
L-AC Guard (SHV connector)
External DC bias voltage: $\pm$ 3000 V
Frequency:
10 kHz to 1 MHz (150 Ω at 10 kHz)
Series capacitance: 110 nF $\pm$ 5%
Input resistance: 100 k Ω $\pm$ 1%

N1261A protection adapter

N1261A-001 protection adapter for HPSMU (triaxial output)

Specifications

Input terminals:
Force (Triaxial)
Sense (Triaxial)
Output terminals:
Force (Triaxial)
Sense (Triaxial)

Supplemental characteristics

Leakage current: 10 pA at 200 V

N1261A-002 protection adapter for GNDU (BNC output)

Specifications

Input terminals:
Force/Sense (Triaxial)
Output terminals:
Force (BNC)
Sense (BNC)

N1261A-003 protection adapter for HPSMU (HV triaxial output)

Specifications

Input terminals:

Force (Triaxial)

Sense (Triaxial)

Output terminals:

Force (HV triaxial)

Sense (HV triaxial)

Supplemental characteristics

Leakage current: 10 pA at 200 V

N1261A-004 protection adapter for GNDU (SHV output)

Specifications

Input terminals:

Force/Sense (Triaxial)

Output terminals:

Force (SHV)

Sense (SHV)

N1262A Resistor Box

N1262A-001 1 M Ω resistor box

Specifications

Input terminals:

HVSMU port, 1 ea. (HV triaxial)

Output terminals:

SHV connector, 1 ea.

Resistance: 1 M Ω $\pm$ 5%

Maximum voltage: $\pm$ 3000 V

Maximum power: 9 W

Supplemental characteristics

Leakage current:

10 pA at 100 V

N1262A-002 100 k Ω resistor box

Specifications

Input terminals:

HVSMU port, 1 ea. (HV triaxial)

Output terminals:

SHV connector, 1 ea.

Resistance: 100 k Ω $\pm$ 5%

Maximum voltage: $\pm$ 3000 V

Maximum power: 6.4 W

Supplemental characteristics

Leakage current: 10 pA at 100 V

N1262A-010 1 k Ω resistor box for gate (triaxial output)

Specifications

Input terminals:

Triaxial connector, 1 ea.

Output terminals:

Triaxial connector, 1 ea.

Resistance: 1 k Ω $\pm$ 10%

Maximum voltage: $\pm$ 200 V

Maximum power: 1 W

Supplemental characteristics

Leakage current: 10 pA at 100 V

N1262A-011 1 k Ω resistor box for gate (SHV output)

Specifications

Input terminals:

Triaxial connector, 1 ea.

Output terminals:

SHV connector, 1 ea.

Resistance: 1 k Ω $\pm$ 10%

Maximum voltage: $\pm$ 200 V

Maximum power: 1 W

Supplemental characteristics

Leakage current: 10 pA at 100 V

Agilent EasyEXPERT Software

Functions

Operation mode:

Application test mode, Classic test mode, Tracer test mode (Curve tracer mode), Quick test mode

Key features:

- Categorized and predefined application library
- Device definition
- Measurement parameter settings
- Save/Recall My Favorite Setups
- Define/customize application library
- Execute measurement (Single/Repeat/Append)
- Quick test execution
- Direct control
- Save/Recall measurement data and settings
- Test result data management
- Import/Export device definition, measurement settings, my favorite setup, measurement data, and application library
- Graph plot display/analysis/printing
- Workspace management
- Self-test, self calibration, diagnostics

Application library

Category:

Sample test definitions for the following applications. They are subject to change without notice.

High Power Device, Utility

Measurement mode details

The Agilent B1505A supports the following measurement modes:

- IV spot
- IV staircase sweep
- IV pulsed spot
- IV pulsed sweep
- IV staircase sweep with pulsed bias
- IV Sampling
- IV High speed sampling
- Multi channel sweep*
- Multi channel pulse spot
- Multi channel pulse sweep
- CV sweep
- C-t sampling
- C-f sweep
- CV (DC bias) staircase sweep
- List sweep
- Linear search**
- Binary search**

* EasyEXPERT supports VAR1 and VAR1'.

** They are supported by FLEX command only.

Each SMU can sweep using VAR1 (primary sweep), VAR2 (secondary sweep), or VAR1' (synchronous sweep).

VAR1

Primary sweep controls the staircase (dc or pulsed) voltage or current sweep.

Maximum number of steps: $N_1=1001$

VAR2

Subordinate linear staircase or linear pulsed sweep. After primary sweep is completed, the VAR2 unit output is incremented.

Maximum number of steps: $N_2=1001$
(condition: $1 \leq N_1 \times N_2 \leq 128128$)

VAR1'

Staircase or pulse sweep synchronized with the VAR1 sweep. Sweep is made with a user specified ratio and offset value. VAR1' output is calculated as $VAR1' = a \times VAR1 + b$, where "a" is the user specified ratio and "b" is the user specified offset value.

CONST

A source unit can be set as a constant voltage or current source depending on the unit.

Sweep measurement time settings:

Hold time:

0 to 655.35 s, 10 ms resolution

Delay time:

0 to 65.5350 s, 100 μ s resolution

(Staircase sweep, multi channel sweep)

0 to 655.350 s, 100 μ s resolution

(CV(DC bias) staircase sweep, AC level sweep, frequency sweep)

Step delay time:

0 to 1 s, 100 μ s resolution

Step output trigger delay time:

0 to (delay time) s, 100 μ s resolution

Step measurement trigger delay time:

0 to 65.535 s, 100 μ s resolution

Staircase sweep measurement mode:

Forces swept voltage or current, and measures DC voltage or current. One channel can sweep current or voltage while up to ten channels can measure current or voltage. A second channel can be synchronized with the primary sweep channel as an additional voltage or current sweep source.

Number of steps: 1 to 1001

Sweep type: linear or logarithmic

Sweep direction:

Single or double sweep

Pulsed sweep measurement mode:

Forces pulsed swept voltage or current, and measures DC voltage or current. A second channel can be programmed to output a staircase sweep voltage or current synchronized with the pulsed sweep output.

Staircase sweep with pulsed bias measurement mode:

Forces swept voltage or current, and measures DC voltage or current. A second channel can be programmed to output a pulsed bias voltage or current. A third channel can be synchronized with the primary sweep channel as an additional voltage or current sweep source.

Sampling (time domain) measurement mode

Displays the time sampled voltage/current data (by SMU) versus time.

Sampling channels: up to 5

Sampling points:

For linear sampling:

1 to 100,001/(number of channels)

For log sampling:

1 to 1 + (number of data for

11 decades)

Sampling mode: linear, log

Sampling interval range:

100 μ s to 2 ms, 10 μ s resolution

2 ms to 65.535 s, 1 ms resolution

For < 2 ms, the interval is $\geq 100 \mu\text{s} + 20 \mu\text{s} \times (\text{num. of channels} - 1)$

Hold time, initial wait time:

-90 ms to -100 μ s, 100 μ s resolution

0 to 655.35 s, 10 ms resolution

Measurement time resolution: 100 μ s

Search measurement mode:

Forces and measures voltage or current by using linear search method or binary search method.

Bias hold function

This function allows you to keep a source active between measurements. The source module will apply the specified bias between measurements when running classic tests inside an application test, in quick test mode, or during a repeated measurement. The function ceases as soon as these conditions end or when a measurement that does not use this function is started.

Current offset cancel

This function subtracts the offset current from the current measurement raw data, and returns the result as the measurement data. This function is used to compensate the error factor (offset current) caused by the measurement path such as the measurement cables, manipulators, or probe card.

Time stamp

The B1505A supports a time stamp function utilizing an internal quartz clock.

Resolution: 100 μ s

Other measurement characteristics

Measurement Control:

Single, Repeat, Append, and Stop

SMU Setting Capabilities:

Limited auto ranging, voltage/current compliance, power compliance, automatic sweep abort functions, self-test, and self-calibration

Arithmetic and analysis functions

User functions

Up to 20 user-defined functions can be defined using arithmetic expressions.

Measured data and pre-defined variables can be used in the computation. The results can be displayed on the LCD.

Arithmetic operators

+, -, *, /, ^, abs (absolute value), at (arc tangent), avg (averaging), cond (conditional evaluation), delta, diff (differential), exp (exponent), integ (integration), lgt (logarithm, base 10), log (logarithm, base e), mavg (moving average), max, min, sqrt, trigonometric function, inverse trigonometric function, and so on

Physical constants

Keyboard constants are stored in memory as follows:

q: Electron Charge, 1.602177 E-19 C

k: Boltzman's Constant, 1.380658 E-23

ϵ (e): Dielectric Constant of Vacuum, 8.854188 E-12

Engineering units

The following unit symbols are also available on the keyboard:

a (10^{-18}), f (10^{-15}), p (10^{-12}), n (10^{-9}), μ or μ (10^{-6}), m (10^{-3}), k (10^3), M (10^6), G (10^9), T (10^{12}), P (10^{15})

Analysis capabilities

Overlay graph comparison

A graphics plot can be stored and overlaid.

Scale

Auto scale and zoom

Marker

Marker to min/max, interpolation, direct marker, and marker skip

Cursor

Direct cursor

Line

Two lines, normal mode, grad mode, tangent mode, and regression mode

Automatic analysis function

On a graphics plot, the markers and lines can be automatically located using the auto analysis setup. Parameters can be automatically determined using automatic analysis, user function, and read out functions.

Data variable display

Up to 20 user-defined parameters can be displayed on the graphics screen.

Analysis functions

Up to 20 user-defined analysis functions can be defined using arithmetic expressions.

Measured data, pre-defined variables, and read out functions can be used in the computation. The results can be displayed on the LCD.

Read out functions

The read out functions are built-in functions for reading various values related to the marker, cursor, or line.

Graph plot

Display mode

Data display window can be printed.

Only X-Y graph can be printed.

Graph plot file

Graph plot can be stored as image data to clip board or mass storage device.

File type: bmp, gif, png, emf

Output

Display modes

X-Y graph, list display, and parameter display

X-Y graph display

X-axis and up to eight Y-axis

Linear and log scale

Real time graph plotting

List display

Measurement data and calculated user function data are listed in conjunction with VAR1 step number or time domain sampling step number. Up to 20 data sets can be displayed.

Other functions

Import/export files.

File type:

Agilent EasyEXPERT format,
XML-SS format, CSV format

Data storage

Hard disk drive, DVD-ROM/CD-R/
CD-RW drive

Interfaces

GPIB, Interlock, USB
(USB 2.0, front 2, rear 2),
LAN (100BASE-TX/10BASE-T),
trigger in/out, digital I/O

Trigger I/O

Only available using GPIB FLEX
commands.

Trigger in/out synchronization
pulses before and after setting and
measuring dc voltage and current.
Arbitrary trigger events can be
masked or activated independently.

Supported external instruments

EasyEXPERT Standard edition:

- Supported by application tests:
4284A/E4980A, 81110A, 3458A

EasyEXPERT Plus edition:

- All external instruments supported
by EasyEXPERT Standard edition

Furnished software

- Prober control execution files
- Desktop EasyEXPERT software
with license-to-use for Standard
edition
- 4155/56 setup file converter tool
(Supported operating systems:
Microsoft® Windows® 2000
Professional, XP Home or
Professional, and Vista Business)

General specifications

Temperature range

Operating: +5 °C to +40 °C

Storage: -20 °C to +60 °C

Humidity range¹

Operating: 20% to 70% RH,
non-condensing

Storage: 10% to 90% RH,
non-condensing

Altitude

Operating: 0 m to 2,000 m (6,561 ft)

Storage: 0 m to 4,600 m (15,092 ft)

Power requirement

ac Voltage: 90 V to 264 V

Line Frequency: 47 Hz to 63 Hz

Maximum volt-amps (VA)

B1505A: 900 VA

Regulatory compliance

EMC:

IEC61326-1:+A1/EN61326-1:+A1
AS/NZS 2064.1

Safety:

CSA C22.2 No.1010.1-1992

IEC61010-1:+A2/EN61010-1:+A2

UL3111-1:1994

Certification

CE, CSA, NRTL/C, C-Tick

Dimensions

B1505A:

420 mm W x 330 mm H x 575 mm D

N1258A module selector:

330 mm W x 120 mm H x 410 mm D

N1259A test fixture:

420 mm W x 272 mm H x 410 mm D

N1260A High voltage bias-tee:

164 mm W x 53 mm H x 125 mm D

N1261A-001 HPSMU protection

adapter (Triaxial output):

80 mm W x 40 mm H x 110 mm D

N1261A-002 GNDU protection adapter

(BNC output):

80 mm W x 40 mm H x 110 mm D

N1261A-003 HPSMU protection

adapter (HV triaxial output):

90 mm W x 40 mm H x 140 mm D

N1261A-004 GNDU protection adapter

(SHV output):

80 mm W x 40 mm H x 125 mm D

N1262A resistor box:

50 mm W x 40 mm H x 125 mm D

Weight

B1505A (empty): 20 kg

B1510A: 2.0 kg

B1512A: 2.1 kg

B1513A: 2.0 kg

B1520A: 1.5 kg

N1258A: 5.0 kg

N1259A: 12.0 kg

N1260A: 0.6 kg

N1261A: 0.3 kg

N1262A: 0.3 kg

Furnished accessories

Measurement cables and adapter

Triaxial cable for HPSMU, 2 ea.

HCSMU cable, 1 ea.

HCSMU Kelvin adapter, 1 ea.

HVSMU cable, 1 ea.

Interlock cable, 1 ea.

Ground unit cable, 1 ea.

Keyboard, 1 ea.

Mouse, 1 ea.

Stylus pen, 1 ea.

Power cable, 1 ea.

Manual CD-ROM, 1 ea.

Desktop EasyEXPERT CD-ROM, 1 ea.

License-to-use for Desktop

EasyEXPERT Standard edition,
1 license

Software CD-ROM

(including utility tools)

Disk set for Agilent

4155B/4155C/4156B/4156C

firmware update, 1 set

SMU number label for the B1505A

installed with SMU, 1 sheet

1. In case of some supplemental characteristics, humidity range is defined as 20% to 50% RH

Order Information

Mainframe and modules

B1505A	Power Device Analyzer/Curve Tracer mainframe
	Configure the following modules:
	High power SMU (HPSMU)
	High current SMU (HCSMU)
	High voltage SMU (HVSMU)
	Multi frequency CMU (MFCMU)
B1505A-015	1.5 m cable
B1505A-030	3.0 m cable
B1505A-050	50 Hz line frequency
B1505A-060	60 Hz line frequency
B1505A-A6J	ANSI Z540 compliant calibration
B1505A-UK6	Commercial calibration certificate with test data
B1505A-ABA	English documentation
B1505A-ABJ	Japanese documentation
B1500A-1CM	Rackmount kit

B1505A accessories

16444A-001	Keyboard
16444A-002	Mouse
16444A-003	Stylus pen
N1253A-100	Digital I/O cable
N1253A-200	Digital I/O BNC box
N1254A-100	GNDU to Kelvin adapter
N1258A	Module selector
N1259A-001	Test fixture
N1259A-010	Inline package socket module (3 pin)
N1259A-011	Universal socket module
N1259A-012	Blank Teflon board
N1259A-020	High voltage bias-tee
N1259A-021	1 MΩ Resistor box
N1259A-022	100 kΩ Resistor box
N1259A-030	1 kΩ Resistor box for gate
N1260A	High voltage bias-tee
N1261A	Protection adapter
N1262A	Resistor box

SMU cables

16493S-001	HCSMU cable (1.5 m)
16493S-002	HCSMU cable (3 m)
16493S-010	HCSMU Kelvin adapter
16493S-020	HCSMU non-Kelvin adapter
16493T-001	High voltage triaxial cable (1.5 m)
16493T-002	High voltage triaxial cable (3 m)
16493U-001	High current BNC cable (1.5 m)
16493U-002	High current BNC cable (3 m)
16494A-001	Triaxial cable (1.5 m)
16494A-002	Triaxial cable (3 m)
16493K-001	Kelvin triaxial cable (1.5 m)
16493K-002	Kelvin triaxial cable (3 m)

CMU accessories

N1300A-001	CMU cable (1.5 m)
N1300A-002	CMU cable (3 m)

Other accessories

16493G	Digital I/O cable
16493J-001	Interlock cable (1.5 m)
16493J-002	Interlock cable (3 m)
16493L-001	GNDU cable (1.5 m)
16493L-002	GNDU cable (3 m)

Additional modules

B1510A	High power source monitor unit retrofit kit
B1512A	High current source monitor unit retrofit kit
B1513A	High voltage source monitor unit retrofit kit
B1520A	Multi frequency capacitance measurement unit retrofit kit



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